

How to Run

Competition Leaderboard System · Group 28 · CS F213 (OOP)

Repository: <https://github.com/thegreatflash/oop-project-group28>

Requirements

- JDK 21 or newer (check with `java -version`)
- Maven is **not** required — the included Maven wrapper (`mvnw`) downloads it automatically on first use

Run locally

```
git clone https://github.com/thegreatflash/oop-project-group28
cd oop-project-group28
./mvnw spring-boot:run
```

Wait for `Started LeaderboardApplication`, then open **`http://localhost:8080`** in a browser. Run the command from the project root so the `data/` folder is found. Stop the server with Ctrl+C (scores are saved on shutdown and restored on the next start).

On Windows, use `mvnw.cmd spring-boot:run` instead of `./mvnw`.

Sign in

Username	Password	Role	Can open
participant1 ... participant3	1234	Participant	Home, Leaderboard
judge1 ... judge3	1234	Judge	+ Judge page (submit scores)
admin	1234	Admin	+ Admin page (reset, export, simulator)

Run in the cloud (no local setup)

1. On the GitHub repository page click **Code** → **Codespaces** → **Create codespace on main** (Java 21 is provisioned automatically).
2. In the Codespace terminal run `./mvnw spring-boot:run`.
3. Click the popup for forwarded port **8080** to open the app in a browser tab.

Run the unit tests

```
./mvnw test
```

Expected result: `Tests run: 20, Failures: 0, Errors: 0 and BUILD SUCCESS`.

Suggested demo flow

1. Sign in as `admin` → **Admin** tab → **Start Simulator** (submits a random judge score about every 0.4 s across all competitions).
2. Open the **Leaderboard** tab and watch the rankings reorder live; the table refreshes automatically every 3 seconds.
3. Switch the **Show** selector to *Top 5* to see the O(k) Top-K view.
4. Sign in as `judge1` in another window and submit a score manually — try an invalid one (e.g. 150) to see validation reject it.
5. On the Admin tab, **Export to CSV** writes the standings to `data/<competition>_leaderboard.csv`.